VALVE SPRINGS

REMOVAL

REMOVAL - CYLINDER HEAD ON

(1) Remove camshafts.
(2) Rotate crankshaft until piston is at TDC on compression.
(3) With air hose attached to adapter tool installed in spark plug hole, apply 90-120 psi air pressure.
(4) Using Special Tool MD-998772-A with adapter 6779 (Fig. 28), compress valve springs and remove valve locks.
(5) Remove valve spring(s).
(6) Remove valve stem seal(s) by a using valve stem seal tool (Fig. 30).

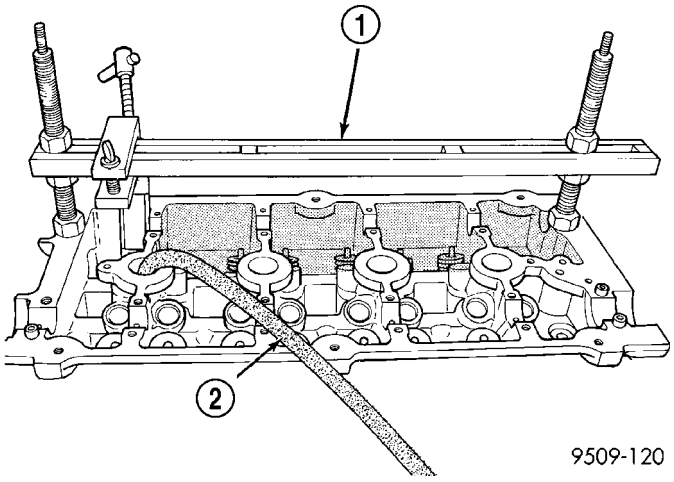

Fig. 28 Valve Spring - Removal/Installation

1 - VALVE SPRING COMPRESSOR MD 998772A
2 - AIR HOSE

REMOVAL - CYLINDER HEAD OFF

(1) With cylinder head removed from cylinder block, compress valve springs using a universal valve spring compressor.
(2) Remove valve retaining locks, valve spring retainers, valve stem seals and valve springs.
(3) Before removing valves, **remove any burrs from valve stem lock grooves to prevent damage to the valve guides.** Identify valves, locks and retainers to insure installation in original location.
(4) Inspect the valves. (Refer to 9 - ENGINE/CYLINDER HEAD/VALVE SPRINGS - INSPECTION)

INSPECTION

(1) Whenever valves have been removed for inspection, reconditioning or replacement, valve springs should be tested for correct tension. Discard the springs that do not meet specifications. The following specifications apply to both intake and exhaust valves springs:

- Valve Closed Nominal Tension - 75 lbs. @ 38.0 mm (1.50 in.)
- Valve Open Nominal Tension - 134 lbs. @ 29.75 mm (1.17 in.)

(2) Inspect each valve spring for squareness with a steel square and surface plate, test springs from both ends. If the spring is more than 1.5 mm (1/16 inch) out of square, install a new spring.

INSTALLATION

INSTALLATION - CYLINDER HEAD ON

(1) Install valve seal/valve spring seat assembly (Fig. 29). Push the assembly down to seat it onto the valve guide.
(2) Install valve spring and retainer, use Special Tool MD-998772-A with adapter 6779 to compress valve springs only enough to install locks. Correct alignment of tool is necessary to avoid nicking valve stems.
(3) Remove air hose and install spark plugs.
(4) Install camshafts and cylinder head cover.

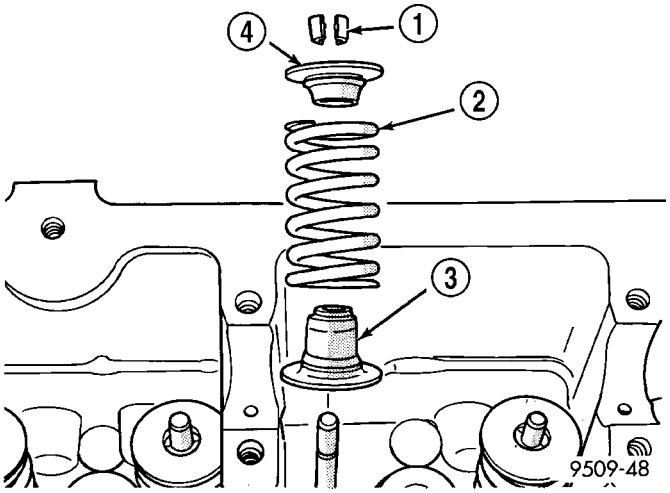

Fig. 29 Valve Stem Seal/Valve Spring Seat - Typical

1 - 3-GROOVE -VALVE RETAINING LOCKS
2 - VALVE SPRING
3 - VALVE SEAL AND VALVE SPRING SEAT ASSEMBLY
4 - VALVE SPRING RETAINER

INSTALLATION - CYLINDER HEAD OFF

(1) Coat valve stems with clean engine oil and insert in cylinder head.
(2) Install new valve stem seals on all valves using a valve stem seal tool (Fig. 30). The valve stem seals should be pushed firmly and squarely over valve guide.

CAUTION: When oversize valves are used, the corresponding oversize valve seal must also be used. Excessive guide wear may result if oversize seals are not used with oversize valves.